

AI-Driven Multimodal Stress and Emotion Analysis with Explainable Intelligence

R26-IT-036

Project Proposal Report

MUNASINGHE M.A.C.D – IT22252586

B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

Department of Information Technology

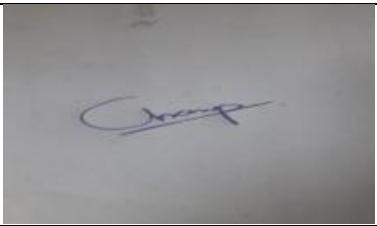
Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
MUNASINGHE M.A.C.D	IT22252586	

The above candidate is conducting research for their undergraduate dissertation under my supervision.

Name	Date	Signature
Prof. Samantha Thelijjagoda		Digitally signed by Samantha Thelijjagoda Date: 2026.03.17 11:10:58 +09'00'

ABSTRACT

Psychological stress has become a growing concern in modern society, particularly among students and online communities where individuals frequently express their experiences and emotional struggles through digital text. Traditional computational approaches for mental health analysis often rely on basic sentiment or emotion classification, which fails to capture deeper psychological factors contributing to stress. According to cognitive behavioral therapy (CBT), stress responses are influenced not only by external stressors but also by internal cognitive distortions such as catastrophizing and overgeneralization. However, many existing natural language processing systems do not clearly distinguish between these external causes and internal thinking patterns, and they often operate as black-box models with limited interpretability. This research proposes an explainable text-mining framework for identifying stressors and CBT-based cognitive distortions from textual data. The study will utilize mental health-related datasets such as Dreaddit and annotated cognitive distortion datasets to train a transformer-based multi-task learning model capable of simultaneously detecting external stressors and internal cognitive distortions. To improve transparency, explainable artificial intelligence techniques will be applied to generate token-level rationale spans that highlight the words or phrases influencing cognitive distortion predictions. Additionally, BERTopic-based topic modeling will be used to discover emerging stress-related themes beyond predefined stressor categories. The system will be implemented using transformer architectures such as RoBERTa and DeBERTa, and is expected to provide accurate identification of stress-related linguistic patterns, interpretable cognitive distortion detection, and discovery of new stress themes, contributing to more transparent and effective AI-driven mental health analysis systems.

Keywords: Stressor Detection, Cognitive Distortions, Explainable AI, Transformer Models

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Senior Professor Samantha Thelijjagoda** and **Senior Professor Junius Anjana** for their valuable guidance, encouragement, and constructive feedback throughout the development of this research proposal. Their expertise and support have been instrumental in shaping this study.

I also wish to thank **Ms. Nuwangi Konara** for providing important psychological insights that helped align the research with mental health perspectives. In addition, I would like to acknowledge the support of the **AIMS (Autonomous Intelligent Machines and Systems) Research Cluster** and the academic staff of the faculty for their guidance and knowledge sharing.

Finally, I would like to thank my research group members for their collaboration and support in developing this project.

TABLE OF CONTENTS

DECLARATION	ii
ABSTRACT.....	iii
ACKNOWLEDGEMENT	iv
TABLE OF CONTENTS.....	v
LIST OF TABLES	vi
LIST OF FIGURES.....	vii
LIST OF APPENDICES	viii
LIST OF ABBREVIATIONS	ix
1. INTRODUCTION	1
1.1 Background and Context	1
1.2 Literature Review	3
1.2 Research Gap	5
2. OBJECTIVE	6
2.1 Main Objective	6
2.2 Sub-Objectives	6
3. Methodology.....	7
4. HIGH-LEVEL SYSTEM ARCHITECTURE	12
5. USER REQUIREMENTS.....	13
5.1 Stakeholder Identification	13
5.2 Functional Requirements	13
5.3 Non-Functional Requirements.....	14
6. COMMERCIALIZATION PLAN	15
7. Budget and Justification.....	18
8. WORK BREAKDOWN STRUCTURE (WBS).....	21
9. GANTT CHART	22
References.....	23
11. APPENDICES	24
Risk Analysis Table.....	24

LIST OF TABLES

Table 1 - User Requirements.....	13
Table 2 - Cost Estimation.....	16
Table 3 - Pricing Strategy	16
Table 4 - Hardware Cost	18
Table 5 - Software Licenses	18
Table 6 - Cloud Services	19
Table 7 - AWS Cloud Hosting.....	19
Table 8 - Human Effort Cost.....	20
Table 9 - Total Estimated Budget	20

LIST OF FIGURES

Figure 1 - High Level Architecture	12
Figure 2 - Work Breakdown Structure.....	21
Figure 3 - Gantt Chart.....	22

LIST OF APPENDICES

Appendix 1 : Risk Analysis Table	25
--	----

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BERTopic	Bidirectional Encoder Representations from Transformers Topic Modeling
CBT	Cognitive Behavioral Therapy
C1–C4	Component 1 to Component 4 (modular pipelines in the system)
DeBERTa	Decoding-enhanced BERT with Disentangled Attention
ECG	Electrocardiogram
EEG	Electroencephalogram
GSR	Galvanic Skin Response
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
LIME	Local Interpretable Model-Agnostic Explanations
NLP	Natural Language Processing
PMC	PubMed Central
RoBERTa	Robustly Optimized BERT Pretraining Approach
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
UMAP	Uniform Manifold Approximation and Projection
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Background and Context

Psychological stress has emerged as one of the most critical mental health challenges globally, with profound effects on students, young adults, and working populations. Increasing academic pressure, social and family expectations, financial difficulties, and workplace demands have significantly contributed to elevated stress levels, which, if unaddressed, can escalate into anxiety, depression, and other mental health disorders [1]. Global health statistics indicate that stress-related mental health conditions continue to rise, placing a growing burden on healthcare systems and society at large. For instance, recent surveys reveal that a substantial percentage of university students experience moderate to severe psychological stress, mainly due to academic workload, uncertainty regarding career opportunities, and social pressures [2]. Similarly, workplace environments have recorded elevated stress levels due to high workloads, tight deadlines, and organizational demands, reflecting the need for effective methods to monitor and understand stress patterns in real-world contexts.

Traditional stress assessment methods, including self-reported questionnaires, clinical interviews, and observational studies, are widely used in psychological research. However, these methods have several critical limitations. Self-reported measures are inherently subjective and retrospective, potentially leading to inaccurate or incomplete reporting of emotional states. Moreover, these approaches are often impractical for continuous monitoring or large-scale stress assessment. With the increasing prevalence of digital communication platforms, individuals now express emotions, thoughts, and stress experiences through online textual communication, such as social media posts, discussion forums, and other digital platforms. These data sources offer a unique opportunity to leverage artificial intelligence (AI) and natural language processing (NLP) techniques to automatically analyze stress patterns and detect mental health signals at scale.

Among the available textual datasets, the Dreaddit dataset developed by Turcan and McKeown [3] has emerged as a benchmark resource for stress detection. It contains thousands of annotated Reddit posts spanning domains such as academic stress, financial challenges, and relationship difficulties. Dreaddit enables the training of AI models capable of identifying linguistic patterns associated with psychological stress. However, most existing NLP methods treat stress detection as a basic sentiment or emotion classification task, focusing on positive versus negative emotional expression without capturing the complex psychological mechanisms behind stress responses.

According to Cognitive Behavioral Therapy (CBT), stress arises not only from external events but also from the way individuals cognitively interpret those events. Cognitive distortions—including catastrophizing, overgeneralization, and negative filtering—are maladaptive thinking patterns that amplify emotional distress. Recent research has explored automated detection of cognitive distortions in textual data using machine learning and NLP techniques, providing

valuable insights into how individuals internally process stressful experiences [4] [1]. Integrating cognitive distortion analysis with stress detection allows for a more comprehensive understanding of psychological responses and enhances the interpretability and clinical relevance of AI systems.

Transformer-based deep learning models, such as RoBERTa and DeBERTa, have demonstrated remarkable performance in natural language understanding tasks, capturing complex contextual relationships within text [4]. However, these models typically function as black boxes, limiting interpretability and raising concerns in sensitive domains such as mental health analysis. Explainable AI (XAI) techniques, including SHAP and LIME, address this challenge by identifying specific words or phrases that strongly influence model predictions, providing token-level explanations for stress and cognitive distortion classification [5] , [6].

Another limitation of many existing stress detection systems is their reliance on predefined categories of stress factors, which can restrict their ability to detect emerging or novel stress patterns. Topic modeling approaches provide a powerful tool for discovering latent themes in large textual datasets. BERTopic, developed by Maarten Grootendorst [7], combines transformer-based embeddings with dimensionality reduction and clustering algorithms to extract semantically meaningful topics. BERTopic has been successfully applied to analyze mental well-being narratives among students and online communities, detecting emerging themes related to academic pressure, emotional challenges, and self-care behaviors [8] [9].

To address these gaps, this research proposes an explainable text-mining framework capable of simultaneously detecting external stressors and internal CBT-based cognitive distortions while generating token-level interpretability for predictions. Additionally, BERTopic-based topic modeling will be employed to identify emerging stress-related themes beyond predefined categories, enabling a deeper understanding of stress patterns expressed in textual communication.

This research aligns with the United Nations Sustainable Development Goal 3 (Good Health and Well-being), which promotes mental health and well-being through innovative technological solutions. Furthermore, the study contributes to the Autonomous Intelligent Machines and Systems (AIMS) research cluster, aiming to develop intelligent systems capable of perception, reasoning, and decision-making through advanced AI techniques.

1.2 Literature Review

Recent advancements in artificial intelligence (AI) and natural language processing (NLP) have enabled researchers to analyze mental health conditions such as stress, anxiety, depression, and emotional distress using computational techniques. Textual data from social media platforms, forums, and online communities have proven particularly valuable for assessing psychological states because users often share personal experiences, emotions, and stressors in these platforms. The ability to automatically analyze these data sources opens possibilities for early detection of stress and identification of cognitive patterns that traditional self-reported surveys or clinical interviews may miss [2] [3].

One of the most widely used datasets for textual stress analysis is the Dreddit dataset, introduced by Turcan and McKeown [3]. Dreddit contains thousands of annotated Reddit posts categorized into domains such as academic stress, financial challenges, relationship issues, and general life stressors. The dataset provides a rich resource for training NLP models to detect stress-related linguistic patterns, including emotion-laden words, expressions of anxiety, and contextually relevant phrases. Researchers have demonstrated that features extracted from textual data, such as narrative structure, syntactic patterns, and semantic associations, can serve as reliable indicators of stress levels, improving the predictive performance of AI-based stress detection systems [2].

While stress detection is well-studied, there has been increasing focus on understanding cognitive distortions irrational or maladaptive thought patterns that intensify emotional distress. Distortions such as catastrophizing, overgeneralization, personalization, and negative filtering play a key role in cognitive behavioral therapy (CBT) and are highly relevant for predicting emotional states [1] [4] [10]. Recent studies have explored automatic detection of cognitive distortions in textual data using machine learning and NLP techniques. For example, transformer-based models like RoBERTa and DeBERTa have been fine-tuned on mental health datasets to detect linguistic patterns indicative of distorted thinking [4]. These approaches provide insights into how individuals internally process stressful experiences, complementing the identification of external stressors such as academic or financial pressures.

Another critical dimension in mental health AI is model interpretability. Deep learning architectures, particularly transformer models, are often criticized for their black-box nature. Predictions made by such models are not easily explainable, limiting their practical utility in sensitive domains like mental health assessment [5] [6]. To overcome this challenge, explainable AI (XAI) techniques, including SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations), have been used to generate token-level interpretations of predictions. These techniques highlight specific words or phrases that strongly influence model outputs, enabling clinicians, researchers, and users to understand the rationale behind stress or cognitive distortion classifications [5]. Token-level explainability ensures that AI models are not only accurate but also transparent, trustworthy, and actionable for mental health interventions.

In addition to classification tasks, topic modeling techniques have emerged as a powerful tool for discovering hidden or emerging themes within large textual datasets. Traditional methods such as Latent Dirichlet Allocation (LDA) are limited in handling short texts and capturing semantic context. To address these limitations, BERTopic, proposed by Maarten Grootendorst integrates transformer-based embeddings, dimensionality reduction, and clustering algorithms to extract semantically meaningful topics from textual corpora. BERTopic has been applied successfully in studies analyzing mental well-being narratives, revealing emerging stress-related themes among university students and online communities, including academic pressures, self-care practices, and social anxiety [11] [10]. This capability to detect latent themes beyond predefined categories is essential for understanding evolving stress patterns in digital communication.

Several studies have combined multiple aspects of stress detection to create multimodal and explainable frameworks. For instance, integrating textual analysis with cognitive distortion detection and explainable AI provides a holistic view of an individual's psychological state, bridging the gap between external stressors and internal cognitive processes [1] [4] [12]. Moreover, studies using BERTopic have shown that topic modeling can complement token-level classification by identifying patterns and trends that may not be captured in labeled datasets, enabling proactive interventions and personalized support [8] [10]

Despite these advances, most existing approaches tend to focus on isolated tasks, such as stress classification, cognitive distortion detection, explainability, or topic discovery, rather than combining all elements in a single, integrated framework. There is a clear need for research that unifies these components into a comprehensive, explainable AI-based text-mining framework capable of simultaneously identifying external stressors, internal cognitive distortions, and emerging stress themes, while providing interpretable predictions at the token level. Such an approach would significantly enhance the reliability, transparency, and clinical relevance of AI-based mental health monitoring systems.

1.3 Research Gap

Despite the significant progress in artificial intelligence and natural language processing for mental health analysis, several critical limitations persist in current systems. One major challenge is the interpretability crisis, commonly known as the “black box” problem. Many transformer-based models, such as RoBERTa and DeBERTa, achieve high accuracy in stress or emotion classification tasks but rarely provide explanations for their predictions [4]. In domains like mental health, this lack of transparency undermines trust among researchers, clinicians, and users, as it is unclear whether the model captures meaningful psychological patterns or merely superficial textual correlations. While explainable AI techniques such as SHAP and LIME have been proposed to highlight influential words or tokens [5,12], their integration into systems that detect both external stressors and cognitive distortions is still limited. Consequently, current methods fail to provide comprehensive and interpretable insights necessary for actionable mental health monitoring.

Another limitation is the unimodal blind spot, where most existing approaches analyze stress using only textual polarity or general sentiment detection [2,3]. These methods typically overlook the distinction between external stressors (e.g., academic or workplace pressures) and internal cognitive distortions (e.g., catastrophizing, overgeneralization) that mediate emotional responses [1,4]. According to cognitive behavioral therapy (CBT), psychological distress arises not just from external events but also from maladaptive cognitive interpretations. Current natural language processing models rarely analyze these two psychological dimensions simultaneously, limiting the ability to extract meaningful mental health insights.

Additionally, most existing frameworks rely on predefined stress categories, restricting their ability to detect novel or evolving stress themes in dynamic online conversations [7,10]. Topic modeling techniques such as BERTopic provide a way to uncover latent themes in large text corpora [6,7], but few studies integrate explainable text classification with emerging theme discovery to create a comprehensive, interpretable system.

Therefore, there is a clear need for a unified, explainable text-mining framework capable of simultaneously identifying external stressors, CBT-based cognitive distortions, and emerging stress themes in textual data while providing transparent token-level explanations [1,2,4,5,7,10,12]. Such a system would enhance interpretability, reliability, and applicability in mental health research and practice.

2. OBJECTIVE

2.1 Main Objective

To develop an explainable text-mining framework that identifies external stressors and Cognitive Behavioral Therapy (CBT) based cognitive distortions from textual data while generating interpretable rationale spans and discovering emerging stress themes using transformer models and BERTopic.

2.2 Sub-Objectives

1. To collect and preprocess mental health text datasets such as Dreddit and cognitive distortion datasets to prepare high-quality data for stress and distortion analysis.
2. To define and organize label categories for stressors and CBT cognitive distortions based on psychological theory and dataset annotations.
3. To develop a transformer-based multi-task learning model capable of simultaneously identifying external stressors and internal cognitive distortions from textual data.
4. To implement explainable artificial intelligence techniques to generate token-level rationale spans that highlight the words or phrases influencing cognitive distortion predictions.
5. To apply BERTopic-based topic modeling to identify emerging stress-related themes beyond predefined stressor categories.
6. To evaluate the performance and interpretability of the proposed model using classification metrics and explanation quality assessment.

3 Methodology

1. Research Approach

This research adopts a **Design Science Research (DSR)** approach combined with an experimental evaluation methodology to design, develop, and assess an explainable artificial intelligence framework capable of identifying stress-related patterns and cognitive distortions from textual data. The design science paradigm is suitable for this study because the research focuses on creating a technological artifact—an intelligent system that integrates advanced natural language processing techniques with explainability mechanisms. The experimental component is used to evaluate the effectiveness of the developed models through systematic testing using benchmark datasets and quantitative evaluation metrics.

The research process follows a structured pipeline consisting of several stages: dataset acquisition, preprocessing and data preparation, model development, explainability integration, topic discovery, system implementation, and performance evaluation. Each stage contributes to the development of a reliable and interpretable AI system capable of analyzing textual indicators of stress and cognitive distortions. The methodology ensures that the research outcome is both scientifically valid and practically applicable for analyzing mental health-related text data.

2. Data Collection and Dataset Preparation

The research utilizes publicly available datasets containing textual content related to stress and cognitive distortions. One of the primary datasets used in this study is the Dreddit dataset, which contains Reddit posts labeled for stress detection. This dataset is widely used in stress detection research and provides annotated examples of stressful and non-stressful textual expressions.

In addition to the Dreddit dataset, annotated cognitive distortion datasets based on Cognitive Behavioral Therapy (CBT) principles are used to identify patterns of distorted thinking. These datasets include textual samples labeled according to different cognitive distortion categories such as overgeneralization, catastrophizing, emotional reasoning, and black-and-white thinking. By combining these datasets, the study enables the system to analyze both external stressors expressed in text and internal cognitive distortions reflected in language patterns.

All datasets used in the research are publicly accessible and anonymized to ensure ethical use of data. The collected data are organized and prepared in structured formats suitable for machine learning model training and evaluation.

3. Data Preprocessing

Before training the models, the textual datasets undergo a comprehensive preprocessing stage to improve data quality and ensure consistency across samples. Natural Language Processing (NLP) preprocessing techniques are applied using Python libraries such as NLTK and Pandas.

The preprocessing pipeline includes several key steps. First, tokenization is performed to break sentences into individual tokens or words, enabling the model to process textual data effectively. Next, text normalization is applied to standardize the format of words by converting text to lowercase and removing unnecessary formatting inconsistencies. The process also includes stop-word removal, which eliminates frequently occurring words such as “the,” “is,” and “and” that contribute little semantic meaning to the analysis.

Additional preprocessing steps involve removing noise and irrelevant textual elements, including URLs, special characters, and excessive punctuation. These steps ensure that the input text focuses on meaningful linguistic patterns relevant to stress and cognitive distortions. The cleaned and normalized data are then transformed into suitable input representations for transformer-based models.

4. Model Development and Architecture

The core analytical component of the research is based on transformer-based deep learning models, particularly the RoBERTa (Robustly Optimized BERT Pretraining Approach) model, which is known for its strong performance in natural language understanding tasks. RoBERTa is used to generate contextual embeddings that capture semantic relationships within text.

The model architecture follows a multi-task learning framework in which a shared transformer encoder processes the textual input and produces contextual feature representations. These representations are then passed to multiple classification heads designed to perform different but related tasks. One classification head is responsible for detecting external stressors expressed in the text, while the second classification head identifies cognitive distortions based on CBT categories.

The multi-task learning approach allows the model to simultaneously learn representations for both tasks while sharing common linguistic features. This improves the efficiency of the learning process and allows the system to capture deeper relationships between stress-related expressions and distorted thinking patterns. The training process involves fine-tuning the pre-trained RoBERTa model on the selected datasets to adapt it to the specific research objectives.

5. Explainable Artificial Intelligence Integration

To ensure transparency and interpretability of the model's predictions, the research integrates Explainable Artificial Intelligence (XAI) techniques. Two widely recognized explainability methods are used in this study: SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations).

SHAP is used to calculate the contribution of individual features or tokens toward the model's prediction, allowing researchers to understand which words or phrases most strongly influence stress or distortion classifications. LIME provides local explanations for individual predictions by approximating the model's behavior around a specific input instance.

By applying these techniques, the system can highlight the words or phrases that contribute most significantly to the classification results. This improves the transparency of the model and enables users to interpret the reasoning behind predictions. In the context of mental health text analysis, explainability is particularly important because it allows researchers and practitioners to better understand how AI systems identify stress patterns and cognitive distortions.

6. Topic Discovery Using BERTopic

In addition to classification tasks, the research incorporates **topic modeling techniques** to identify underlying themes in stress-related discussions. The study utilizes **BERTopic**, a modern topic modeling framework that combines transformer embeddings with clustering techniques to discover meaningful topics within textual data.

BERTopic works by first generating semantic embeddings using transformer models and then applying dimensionality reduction and clustering algorithms to group similar text samples together. This process enables the identification of recurring themes in stress-related conversations, such as academic pressure, financial stress, social isolation, or work-related challenges.

The integration of topic modeling allows the research to go beyond predefined labels and explore hidden patterns within the dataset. This exploratory analysis provides deeper insights into the types of stressors discussed in textual data and enhances the interpretability of the overall system.

7. System Implementation

The implementation of the proposed framework is carried out using Python as the primary programming language, due to its extensive support for machine learning and natural language processing tasks. The development environment used for experimentation and model

development is Visual Studio Code, which provides an efficient environment for coding, debugging, and model experimentation.

Several libraries and frameworks are utilized throughout the implementation process. Hugging Face Transformers is used to access and fine-tune pre-trained transformer models such as RoBERTa. NLTK and Pandas are employed for data preprocessing and analysis. For model experimentation and interactive analysis, tools such as Jupyter Notebook or similar environments may also be used.

To improve reproducibility and deployment capabilities, the system environment is containerized using Docker, and images may be managed through Docker Hub. Containerization ensures that the developed system can be easily replicated and executed across different environments without dependency conflicts.

8. Model Evaluation and Validation Metrics

The effectiveness of the developed models is evaluated using several standard machine learning evaluation metrics. These include accuracy, precision, recall, and F1-score, which collectively measure the performance of the classification models. Accuracy measures the overall proportion of correct predictions, while precision and recall evaluate the model's ability to correctly identify relevant stress or distortion categories.

The F1-score is used as a balanced metric that considers both precision and recall, providing a more reliable evaluation in cases where class distributions may be imbalanced. In addition to these metrics, the evaluation process may include cross-validation techniques to ensure that the model generalizes well to unseen data.

The evaluation of explainability components involves analyzing whether the highlighted rationale segments correspond to meaningful cognitive distortion indicators identified in the annotated datasets. This validation step ensures that the model's explanations are not only technically correct but also meaningful from a psychological perspective.

9. Ethical Considerations

Since the research involves analyzing text related to stress and mental health, ethical considerations play an important role in the methodology. All datasets used in the study are publicly available and anonymized, ensuring that no personally identifiable information is included in the analysis.

The research adheres to ethical guidelines related to data privacy, responsible AI development, and respectful handling of mental health-related content. The system is intended for research

and analytical purposes rather than clinical diagnosis, and its outputs are designed to support understanding of stress patterns rather than replace professional mental health assessments.

By ensuring responsible data usage and transparent model behavior, the research aims to contribute to the development of ethical and trustworthy AI systems for analyzing psychological and emotional patterns in textual data.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

C3-High Level Architecture for Stress Themes via Rationale-Based Text Mining

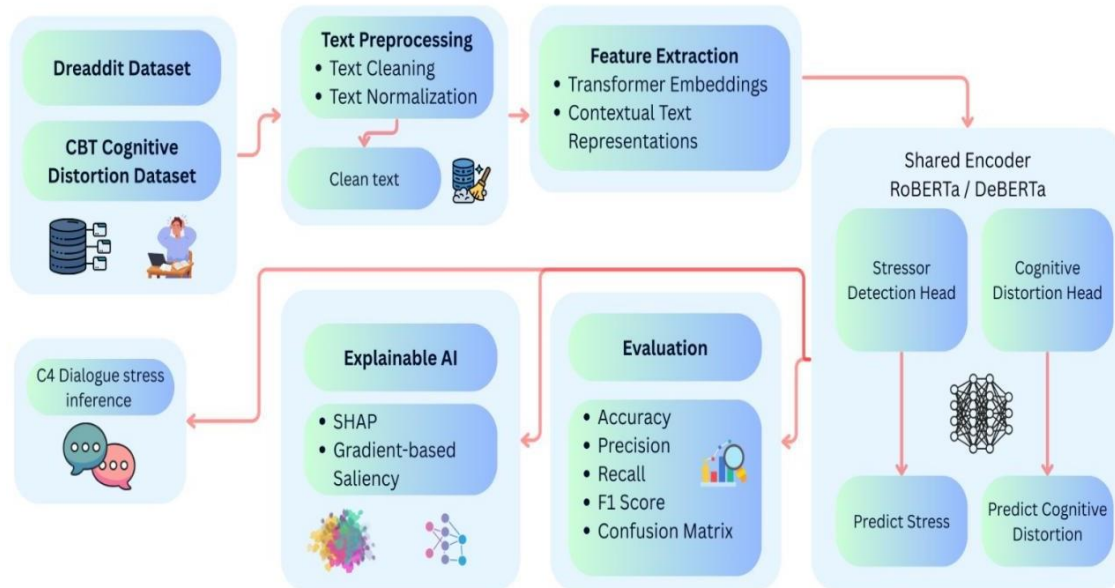


Figure 1 - High Level Architecture

This architecture shows the workflow of the proposed system for detecting stressors and cognitive distortions from text. After preprocessing the datasets, transformer embeddings are extracted and processed by a shared RoBERTa/DeBERTa model with two classification heads. Explainable AI techniques provide interpretation of predictions, and the system performance is evaluated using standard classification metrics.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The following stakeholders interact with or benefit from the system:

Stakeholder	Role
Students / Users	Provide text inputs and receive stress analysis results.
Mental Health Professionals	Use insights to understand stress patterns and cognitive distortions.
System Administrators	Manage system deployment, monitoring, and maintenance.
Researchers / Developers	Improve the model and evaluate system performance.

Table 1 - User Requirements

5.2 Functional Requirements

The system must provide the following functionalities:

1. Stress analysis will enable users to input text data in the system.
2. The system will recognize stress-related categories (e.g., academic, financial, relationship).
3. The system will identify such cognitive distortions as the catastrophizing and overgeneralization (based on CBT).
4. The system will emphasize significant words or phrases that affected the prediction.
5. The system will produce explainable results based on algorithms like SHAP or gradient-based saliency.
6. The system will give probabilities of prediction and classification labels of the observed stressors and distortions.
7. The system will enable a researcher to conduct topic discovery with the use of BERTopic to discover some stress themes.
8. The system will archive the analysis findings to be evaluated and researched on.

5.3 Non-Functional Requirements

- Performance

Within 25 seconds to five seconds the system should handle a user input of text and give results.

- Scalability

Cloud-based infrastructure should provide the support of more than one user concurrently.

- Security

To ensure privacy, user information and written communications should be safely stored and analyzed.

- Usability

The system must offer an easy web-based interface where individuals can easily post text and get results.

- Reliability

The system must have a steady performance and offer predictability as it is used repeatedly.

- Maintainability

The system must be made in a modular architecture in such a way that the individual components can be enhanced separately.

6. COMMERCIALIZATION PLAN

Customer Persona and Target Market

The proposed system is designed for mental health technology platforms, educational institutions, and digital well-being applications that aim to monitor stress and cognitive patterns in users. Key potential customers include: higher education institutions and schools seeking early detection of student stress; mental health assistance platforms offering AI-powered counseling; technology companies developing digital therapeutics for healthcare; and organizations implementing employee well-being monitoring systems. The typical customer profile consists of mental health professionals, educational administrators, and HR managers who require intelligent tools to identify patterns of psychological stress and cognitive distortions in textual communication, enabling early intervention and better mental health support.

Value Proposition

The system delivers an explainable AI-based text analysis framework capable of detecting both external stressors (e.g., academic pressure, financial concerns, relationship issues) and internal cognitive distortions (e.g., catastrophizing, overgeneralization). Unlike conventional sentiment analysis tools that classify text as merely positive or negative, this system provides psychologically meaningful insights with clear rationale explanations, highlighting the specific textual elements driving each prediction. This transparency enhances trust, reliability, and applicability for both professionals and end-users, making it suitable for use in clinical, educational, and organizational mental health contexts.

Revenue Model

The system can be commercialized using a Software-as-a-Service (SaaS) model, with revenue streams including subscription fees for mental health platforms; licensing agreements with universities and healthcare institutions; and per-use charges for text analysis services. Example pricing could be structured as follows: an Essential Plan at \$10 per month per organization offering basic text analysis, and a Professional Plan at \$30 per month per organization providing advanced analytics, interpretability reports, and emerging stress-topic discovery. This model ensures scalability, affordability, and accessibility while maintaining continuous revenue generation for system maintenance and enhancement.

Cost Estimation

The operational cost mainly includes cloud hosting and maintenance.

Estimated monthly operational costs:

Cost Component	Estimated Cost
AWS EC2 hosting	\$10 – \$15
AWS S3 storage	\$1 – \$2
Monitoring & API services	\$2 – \$3

Table 2 - Cost Estimation

Pricing Strategy

A tiered pricing strategy will be adopted to attract different types of users.

Example pricing structure:

Plan	Features	Price
Basic	Stress detection API	\$10/month
Professional	Stress + cognitive distortion detection	\$30/month
Enterprise	Full analytics dashboard and integration support	Custom pricing

Table 3 - Pricing Strategy

This strategy allows both small organizations and large institutions to adopt the platform.

Competitive Advantage

The proposed system has a few benefits over the current solutions: Dual-head transformer architecture which identifies stressors and cognitive distortions. Explainable AI methods like SHAP and gradient-based saliency to reveal significant words that affect the predictions. BERTopic topic discovery, which allows discovering emerging themes of stress. CBT theory based psychologically oriented interpretation. These personalities give more valuable mental health information than conventional sentiment analysis tools.

Intellectual Property Concerns

The intellectual property that can be derived out of this research can be: The model diagram of stressor detection and cognitive distortion analysis. The psychological reasoning explainable AI interpretation pipeline. The combination of topic modeling and the cognitive distortion detection. The project code and trained models can be safeguarded using copyright or software licenses, and commercialization can be done by using API based access control.

7. Budget and Justification

1. Hardware Cost

Item	Estimated Cost	Justification
Personal Laptop	\$0	Used for coding, preprocessing data, and running lightweight experiments.

Table 4 - Hardware Cost

The project does not require specialized hardware because most heavy computations are executed on Google Colab GPUs.

2. Software Licenses

Software	Cost	Justification
Python	\$0	Main programming language for the project.
PyTorch / HuggingFace Transformers	\$0	Used for fine-tuning RoBERTa / DeBERTa models.
Scikit-learn	\$0	Used for evaluation metrics and preprocessing.
BERTopic	\$0	Used to discover new stress-related themes in text.
SHAP / Captum	\$0	Used for Explainable AI analysis.

Table 5 - Software Licenses

All software tools used in this project are open-source; therefore, no licensing cost is required.

3. Cloud Services

Google Colab (Model Training)

Service	Estimated Cost	Justification
Google Colab Free	\$0	Used for training NLP models using GPU resources.
Google Colab Pro (Optional)	~\$10/month	Provides longer runtime and faster GPU for transformer fine-tuning.

Table 6 - Cloud Services

4 .AWS Cloud Hosting (Deployment)

Service	Estimated Cost	Justification
AWS EC2 (t3. micro)	~\$8–\$12/month	Used to host the backend API and model inference service.
AWS S3	~\$1–\$2/month	Used for storing datasets, model files, and logs.
AWS API Gateway / Load Balancing (optional)	~\$1–\$3/month	Used to expose the API for the web application interface.

Table 7 - AWS Cloud Hosting

Estimated AWS hosting cost:

~\$10 – \$20 per month. If AWS Free Tier is used, the cost may be \$0 during the first year.

5. Human Effort Cost

Role	Cost
Student Research Work	\$0

Table 8 - Human Effort Cost

The system is developed as part of a final-year undergraduate research project, so human labor cost is not included.

6. Total Estimated Budget

Category	Monthly (LKR)	8-Month (LKR)	8-Month (USD)
Cloud Hosting (AWS)	4,600 – 6,200	36,800 – 49,600	\$118 – \$160
Model Training (Colab Pro)	3,111	24,888	\$80
Travel & Transport	5,000 – 7,500	40,000 – 60,000	\$128 – \$192
Internet & Data	3,500	28,000	\$90
Printing & Stationery	—	3,000	\$10

Table 9 - Total Estimated Budget

Estimated Total Monthly Cost: Rs. 15,000.00 – Rs. 17,000.00

8. WORK BREAKDOWN STRUCTURE (WBS)

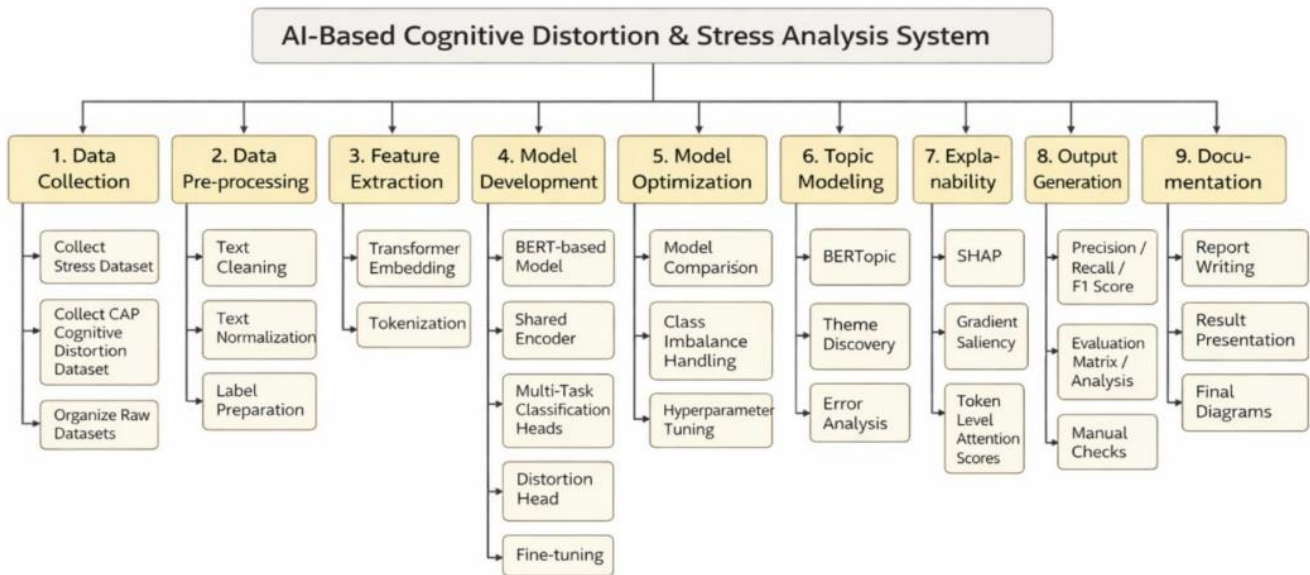


Figure 2 - Work Breakdown Structure

9. GANTT CHART

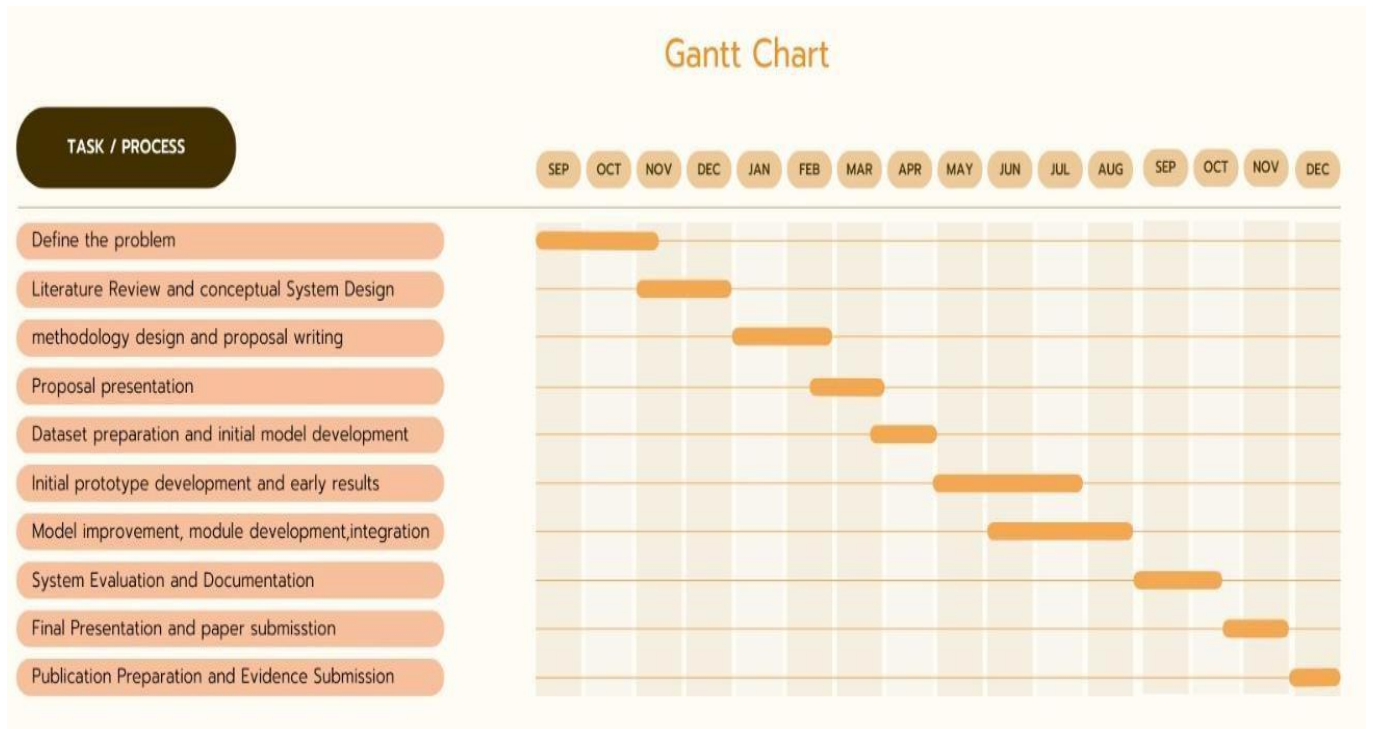


Figure 3 - Gantt Chart

References

- [1] S. Archie , K. Jeroen and Y. Helen , "A Survey of Cognitive Distortion Detection and Classification in NLP," 2025.
- [2] S. Benjamin, S. Scott, H. Martin, B. Sherry and R. Parisa, "Automatic Detection and Classification of Cognitive Distortions in Mental Health Text," *2020 IEEE 20th International Conference on Bioinformatics and Bioengineering (BIBE)*, pp. 275-280, 2020.
- [3] W. Bichen, Z. Yanyan , L. Xin and Q. Bing , "Cognitive distortion based explainable depression detection and analysis technologies for the adolescent internet users on social media," 17 January 2023.
- [4] K. Crane, "US Mental Health Statistics 2025: Essential Facts & Trends," *South Denver Therapy*, 2025.
- [5] T. Elsbeth and M. Kathy , "Dreaddit: A Reddit Dataset for Stress Analysis in Social Media," *Proceedings of the Tenth International Workshop on Health Text Mining and Information Analysis (LOUHI 2019)*, 2019.
- [6] N. Tanya, A. Girija and T. Ananthakrishna , "Stress detection using natural language processing and machine learning over social interactions," *Journal of Big Data*, vol. 9, 2022.
- [7] G. Maarten , "BERTopic: Neural topic modeling with a class-based TF-IDF procedure," 11 March 2022.
- [8] L. Kamini , R. Shalli and S. Dr. Mohammad , "Explainable machine learning for mental health prediction from social media behavior: a nested cross-validation study with SHAP and LIME interpretability," *Discover Mental Health*, vol. 6, no. 1, January 2026.
- [9] M. A. Noor Ul , N. Sanam , J. M. Muhammad and A. A. Syed, "Explainable Machine Learning for Mental Health Detection Using NLP," vol. 14, 2025.
- [10] T. Pharmacally, "Global Mental Health in 2025: Evidence, Gaps, and the Way Forward-Analysis of WHO Reports," 09 September 2025.
- [11] A. Mahmoud , B. v. Niels , T. Benjamin, P. Ville , O. Jonas , Y. Koji and H. Simo, "Investigating mental wellbeing self-care in higher education using BERTopic modeling," *PMC*, vol. 5, no. 1, 23 Nov 2025.
- [12] T. S. Justin, L. Kevin , D. Xiruo , C. Ayesha, H. J. William, C. Trevor and -Z. B. Dror , "Automated Detection of Cognitive Distortions in Text Exchanges Between Clinicians and People With Serious Mental Illness," *Psychiatr Serv*, vol. 74, pp. 407-410., 1 Apr 2023.

11. APPENDICES

Risk Analysis Table

Risk ID	Risk Description	Impact	Probability	Mitigation Strategy
R1	Insufficient or imbalanced dataset for stressor and cognitive distortion detection	High	Medium	Use multiple datasets such as Dreddit and cognitive distortion datasets; apply data augmentation and preprocessing techniques.
R2	Transformer models (RoBERTa / DeBERTa) require high computational resources	Medium	Medium	Use Google Colab GPU resources and optimize model training with batch size tuning and early stopping.
R3	Explainable AI methods (SHAP, LIME) may increase computational overhead	Medium	Medium	Apply explanation techniques only during inference or analysis phase rather than during full training.
R4	Difficulty in accurately detecting cognitive distortions due to subtle language patterns	High	Medium	Fine-tune transformer models carefully and evaluate using multiple metrics such as precision, recall, and F1-score.
R5	Topic modeling may produce irrelevant or unclear topics	Medium	Medium	Optimize BERTopic parameters and validate discovered topics with manual inspection.
R6	Ethical concerns when analyzing mental health data	High	Low	Use publicly available anonymized datasets and follow ethical research guidelines.
R7	Model predictions may be misinterpreted by non-expert users	Medium	Medium	Provide clear explanations using SHAP/LIME and include confidence scores with predictions.

R8	Deployment issues when integrating the system with cloud infrastructure	Medium	Low	Use Docker containerization and test deployment in controlled environments before production.
R9	Limited system scalability when multiple users access the system simultaneously	Medium	Low	Deploy using scalable cloud services such as AWS EC2 with load balancing if required.
R10	Time constraints during research development and experimentation	Medium	Medium	Follow a structured Gantt chart and iterative development approach.

Table 10 - Risk Analysis Table